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E-commerce Dashboard – Sales Data Analysis - KPI & Overall Sale

Project goal

The project's goal was to create an interactive dashboard in Power BI based on sales data. The analysis included key sales metrics, discounts, payment methods, customer locations, and demographic structure.

Data source

Data downloaded from Kaggle(CSV). This dataset contains 55,000 entries

About the Dataset:

- CID (Customer ID): A unique identifier for each customer.
- TID (Transaction ID): A unique identifier for each transaction.
- Gender: The gender of the customer, categorized as Male or Female.
- Age Group: Age group of the customer, divided into several ranges.
- Purchase Date: The timestamp of when the transaction took place.
- Product Category: The category of the product purchased, such as Electronics, Apparel, etc.
- Discount Availed: Indicates whether the customer availed any discount (Yes/No).
- Discount Name: Name of the discount applied (e.g., FESTIVE50).
- Discount Amount (INR): The amount of discount availed by the customer.

- Gross Amount: The total amount before applying any discount.
- Net Amount: The final amount after applying the discount.
- Purchase Method: The payment method used (e.g., Credit Card, Debit Card, etc.).
- Location: The city where the purchase took place.

Stages of working with data

Data import and preparation

- Importing data into Power BI.
- Rearranging columns and changing data types
- Verification of missing data (null values):Single empty values in Discount Name when there was no discount - left because they are logical.
- Checking the uniqueness of customer IDs (CID) and transactions (TID) → no duplicates.
- Adding auxiliary columns:
  - ➔ Splitting the Purchase Date column into several time columns (Year, Month, Quarter) → better filter performance and faster chart creation.
  - ➔ Transaction amount variance (difference between gross, discount and net).
- Creating a table with unique customers (VALUES on CID).
- Ordering months by month number (to sort charts correctly).

Measures (DAX)

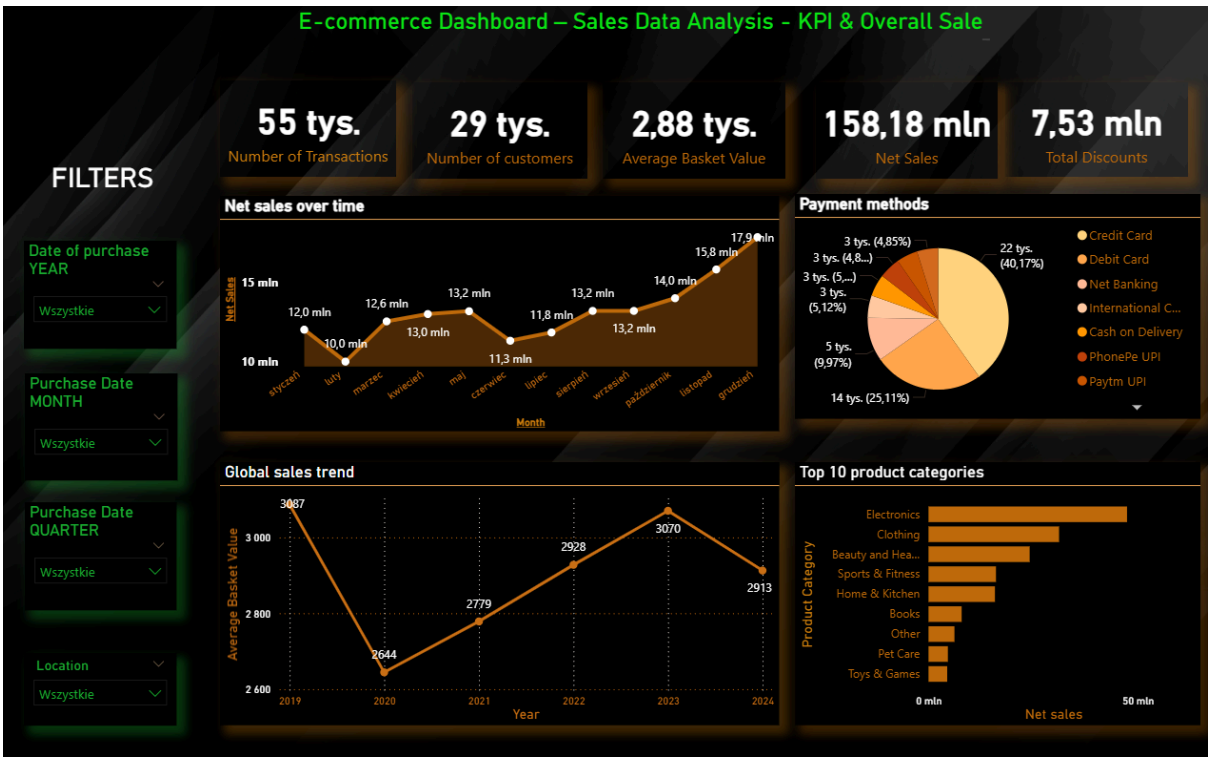
- Average Basket Value = AVERAGE('Table'[Net Amount])
- Net Sales = SUM('Table'[Net Amount])
- Number of Transactions = COUNTROWS('Table')
- Number of customers = DISTINCTCOUNT('Table'[CID])
- SaleWithoutDiscount = CALCULATE(SUM('Table'[Net Amount]), 'Table'[Discount Availed] = "No")
- SaleWithDiscount = CALCULATE(SUM('Table'[Net Amount]), 'Table'[Discount Availed] = "Yes")
- Total Discounts = SUM('Table'[Discount Amount])
- PercentWithDiscount = DIVIDE([SaleWithDiscount], [Net Sales],0)

The dashboard has been divided into 6 thematic cards to facilitate the analysis of individual areas (KPIs, discounts, locations, demographics, payment methods).Filters enable detailed analysis, e.g. for a specific location or age group.Consistent colors and readability of charts have been maintained.

Description of individual dashboard cards

1 card KPI & Overall Sales

- KPI: net sales, number of transactions, number of customers, total discounts, average basket value.
- Net sales trend over time (by year).
- Top 10 product categories by net sales.
- Share of payment methods.
- Global sales trend.
- Filters: year, month, quarter, location.



Conclusions

55,000 transactions and 29,000 customers – on average, each customer made nearly 2 purchases, indicating a recurring customer base rather than one-time purchases.

The average basket value was 2.88 thousand – a stable and relatively high amount for a single order.Total net sales reached 158.18 million, while total discounts granted amounted to 7.53 million, meaning that discounts account for approximately 4.8% of sales value.The sales trend shows dynamic growth between 2019 and 2023, peaking in 2023 at 15.8 million. In 2024, a decline to 17.9 million is visible (note: in the new data, the decline is less significant), which may still require analysis of potential causes, such as seasonality, changes in marketing strategy, or customer churn.

The average basket value decreased in 2020 (likely due to the impact of the pandemic) but gradually increased until 2023 (reaching 3,074). In 2024, the average basket value slightly decreased (2,913), despite a general increase in sales – this suggests that the sales volume remained stable, but individual orders became slightly smaller.

Credit card (40%) remains the dominant payment method.

Electronics and Clothing are the top-performing product categories, generating over 50% of total sales.

Initial business diagnosis:

- The company should focus on restoring the dynamic growth trend observed in 2022–2023, analyzing the reasons behind the sales stabilization or slower growth in 2024 (e.g., market saturation, seasonality, or decreased marketing activity).

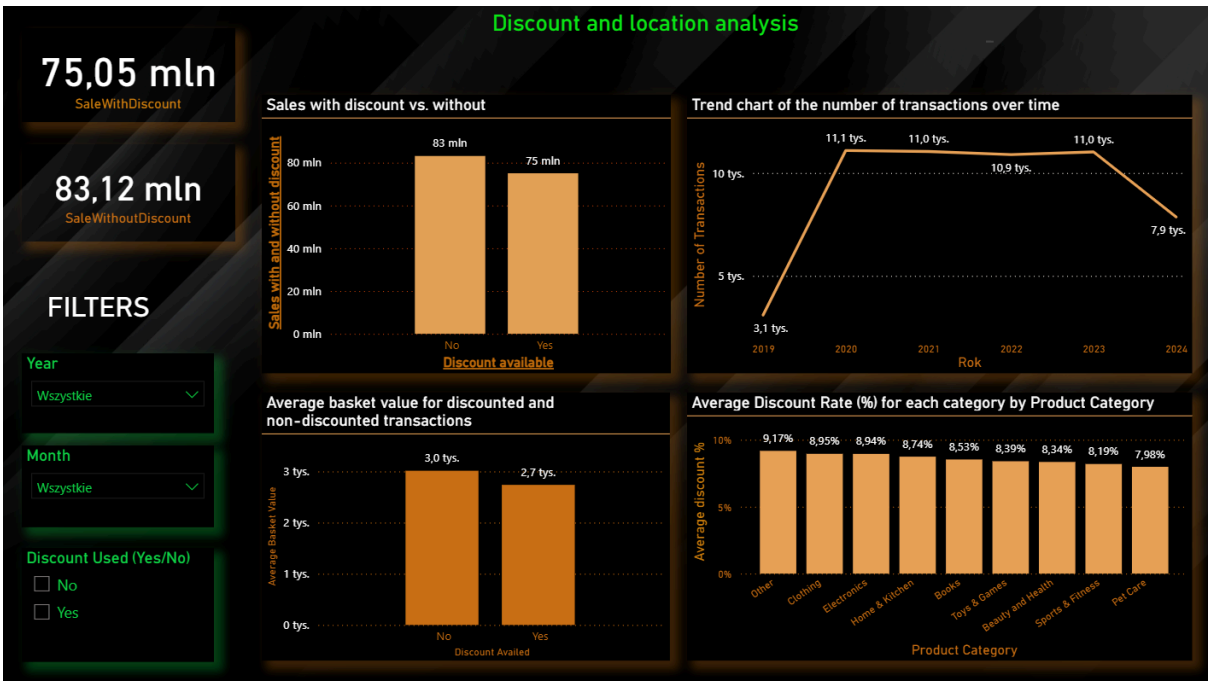
- It is worth monitoring changes in payment methods, especially the growing popularity of UPI and mobile payment solutions, which may gain market share in the coming years.

- Promotional campaigns could be targeted at less popular product categories, such as Books and Pet Care, to diversify revenue sources and increase market share in these segments.

- In addition, the company may explore retention strategies, since the average number of transactions per customer suggests repeat purchases, but there is still room to increase customer loyalty.

Card 2. Discount and location analysis

- Comparison of sales with and without discount.
- Average discount by product category.
- Trend in the number of transactions over time.
- Net sales by location.
- Filters: discount usage (Yes/No), year, month.



Conclusions

Transactions without a discount generate higher net sales – 83.12 million vs 75.05 million for transactions with a discount. This means that non-discounted purchases account for a greater share of revenue, despite the attractiveness of discounts.

The number of transactions increased sharply from 2019 (3.1 thousand) to 2020 (11.1 thousand), and then stabilized at around 11 thousand per year until 2023. In 2024, there was a decline to 7.9 thousand transactions, which may require corrective actions (e.g., marketing campaigns, new discount strategies).

Discounts positively influence the number of transactions, but reduce the average basket value: 3.0 thousand without a discount vs 2.7 thousand with a discount, which is a typical promotional effect.

Among product categories, Pet Care still has the lowest average discount rate (7.98%), indicating lower price competition or stable demand without promotions. Filters allow you to analyze trends by year, month, and discount usage.

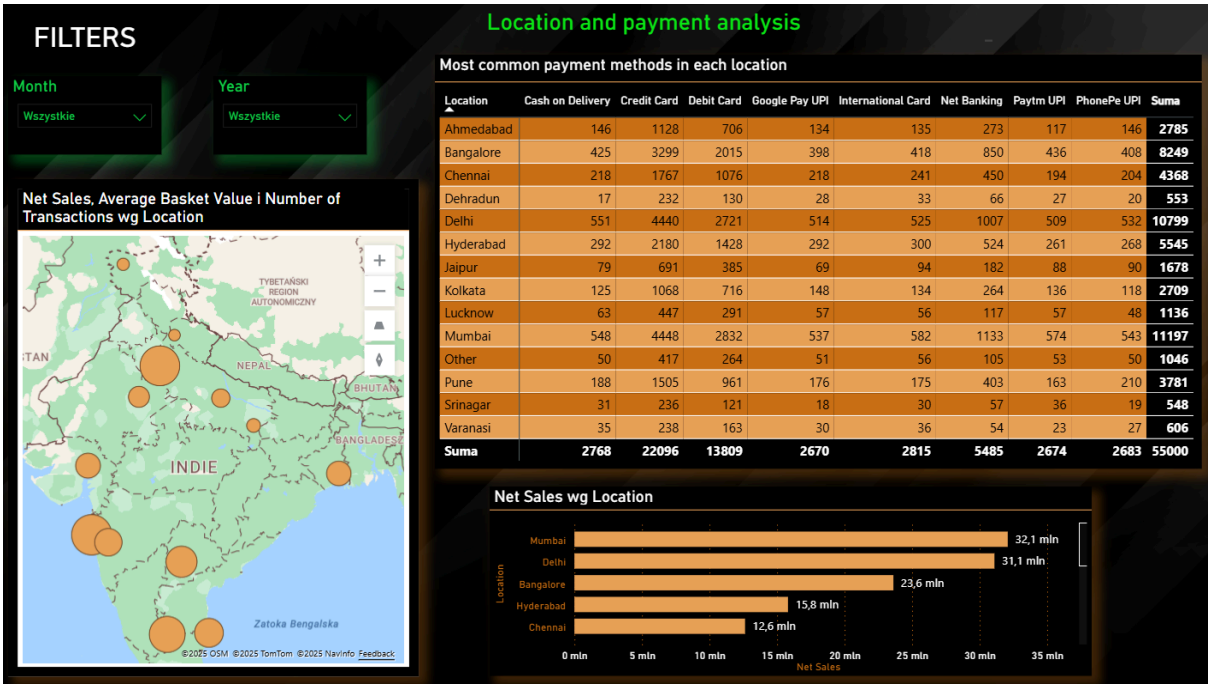
Initial business diagnosis:

- Discounts drive sales volume but reduce the value of individual transactions.

- It's recommended to conduct a profitability analysis of promotions in categories with the highest discounts, to ensure they generate expected margins.
- The drop in the number of transactions in 2024 suggests the need for additional marketing, promotional, or loyalty-building activities, especially to stabilize sales after previous years of growth.

Card 3. Location and payment analysis

- Map of net sales and transactions by location.
- Location ranking by sales.
- A matrix/chart showing the popularity of payment methods in different cities.
- Filters: location, product category.



Most common payment methods in each location

Location	Cash on Delivery	Credit Card	Debit Card	Google Pay UPI	International Card	Net Banking	Paytm UPI	PhonePe UPI	Suma
Ahmedabad	146	1128	706	134	135	273	117	146	2785
Bangalore	425	3299	2015	398	418	850	436	408	8249
Chennai	218	1767	1076	218	241	450	194	204	4368
Dehradun	17	232	130	28	33	66	27	20	553
Delhi	551	4440	2721	514	525	1007	509	532	10799
Hyderabad	292	2180	1428	292	300	524	261	268	5545
Jaipur	79	691	385	69	94	182	88	90	1678
Kolkata	125	1068	716	148	134	264	136	118	2709
Lucknow	63	447	291	57	56	117	57	48	1136
Mumbai	548	4448	2832	537	582	1133	574	543	11197
Other	50	417	264	51	56	105	53	50	1046
Pune	188	1505	961	176	175	403	163	210	3781
Srinagar	31	236	121	18	30	57	36	19	548
Varanasi	35	238	163	30	36	54	23	27	606
Suma	**2768**	**22096**	**13809**	**2670**	**2815**	**5485**	**2674**	**2683**	**55000**

Conclusions

Highest net sales are in Mumbai (32.1 mln), Delhi (31.1 mln), and Bangalore (23.6 mln) – these three locations account for the majority of the company's sales.

Hyderabad (15.8 mln) and Chennai (12.6 mln) also generate significant revenue, confirming their importance on the Indian e-commerce market.

Credit Card dominates as the most popular payment method in all locations. However, in Bangalore and Hyderabad, Debit Card and UPI payments (Google Pay, PhonePe) have a slightly higher share compared to other cities.

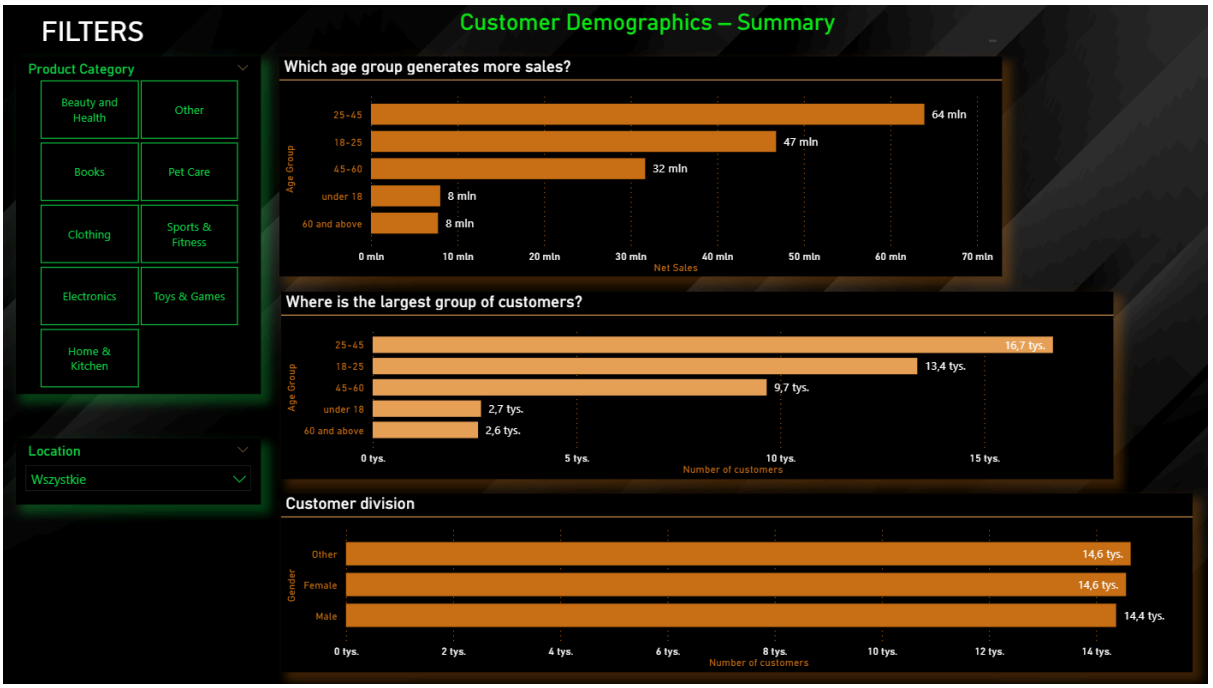
The map confirms that sales are concentrated in major urban areas, which is typical for online sales in India.
Filters allow the analysis of payment preferences and sales by month and year.

Initial business diagnosis:

- The company should maintain and strengthen its position in key cities like Mumbai, Delhi, and Bangalore.
- It is worth considering initiatives to promote alternative payment methods (e.g., UPI) in both large and smaller cities.
- There is potential for growth in less developed locations, such as Jaipur, Varanasi, Srinagar, where the sales volume and number of transactions are still low.

Card 4. Customer Demographics – Summary

- Which age group generates more sales.
- Where is the largest group of customers.
- Customer division
- Filters- Product Category, Location.To check which segments are dominant in particular categories.



Conclusions

The 25–45 age group generates the highest net sales – 64 million, making it the most important customer segment.
The 18–25 age group contributes 47 million, also representing a significant part of the sales.
The older groups (45–60, under 18, and 60+) each generate approx. 8 million, indicating a much smaller contribution compared to younger segments.
In terms of customer numbers:

- 25–45 years: 16.7 thousand customers.
- 18–25 years: 13.4 thousand customers.
- 45–60 years: 9.7 thousand customers.
- Younger and older customers (under 18 and 60+) represent smaller segments (approx. 2.6–2.7 thousand each).

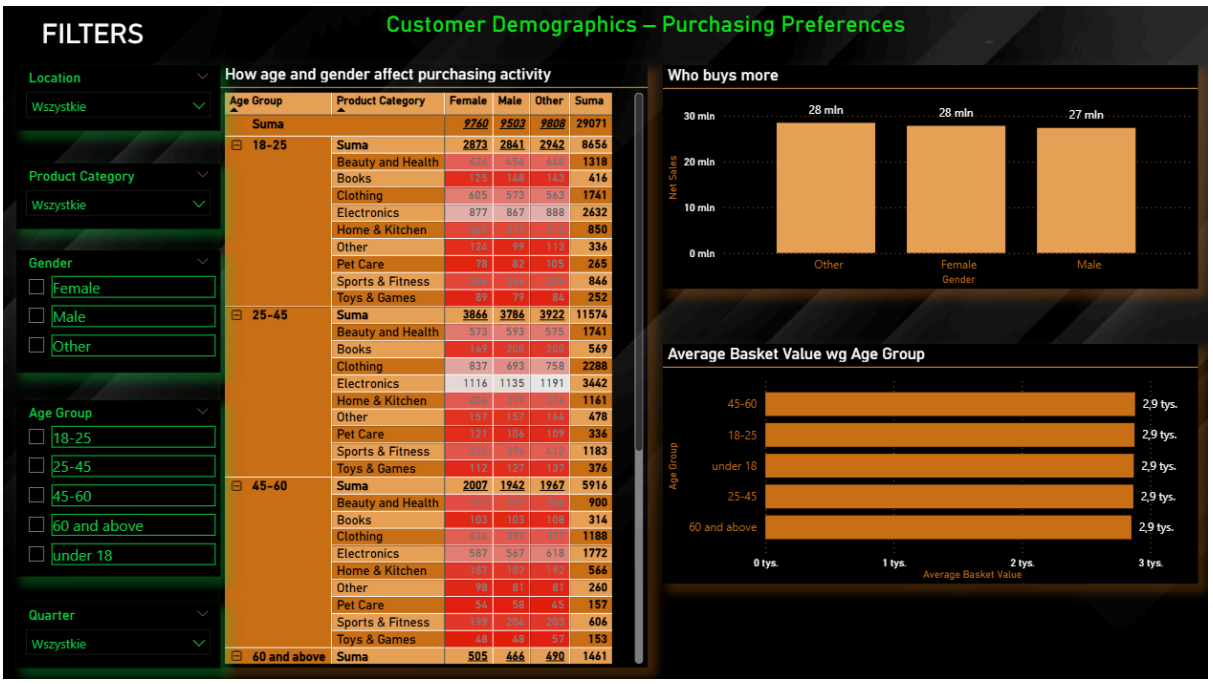
The gender distribution is balanced, with Female: 14.6 tys., Male: 14.4 tys., Other: 14.6 tys., meaning the offer appeals equally to all gender groups.

Initial business diagnosis:

- The main source of revenue is still the working-age group (25–45 years), which is typical for e-commerce.
- There is potential for growth in the 45+ and under-18 segments, although their sales are currently low compared to younger groups.
- The gender balance shows no need for targeting one gender over another, which allows marketing campaigns to be inclusive.

Card 5. Customer Demographics – Purchasing Preferences

- How age and gender affect purchasing activity.
- Who buys more.
- Average Basket Value wg Age Group.
- Filters- Location, Product Category, Gender,Age Group,Quarter



Conclusions

The most active customer groups are:

- 18–25 years: 16,431 transactions.
- 25–45 years: 22,010 transactions.

These two groups together account for the majority of all transactions (almost 70% of total activity).
The most frequently purchased categories are still: Electronics, Clothing, Beauty and Health.
Gender differences:

- Female, Male, and Other customers have very similar purchasing activity in total (each group over 18,000 transactions).
- Some categories show clear preferences: e.g., Beauty and Health is more popular among female customers, while Electronics and Clothing are evenly distributed.

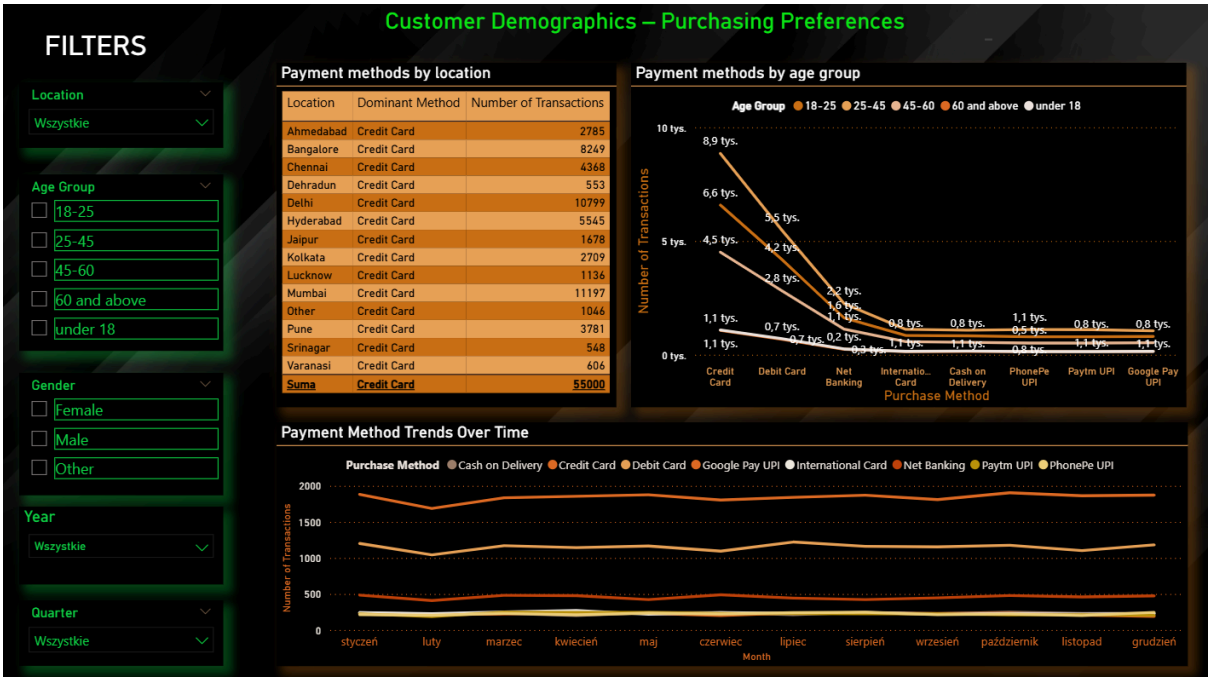
Average Basket Value remains stable across age groups (~2.9 tys.), indicating that the age group influences the number of purchases, not the amount spent per transaction.

Initial business diagnosis:

- The key purchasing power lies within the 18–45 years old segment – they drive most of the transactions and revenue.
- Differences between age groups are mainly visible in the purchase frequency, not in basket size.
- It is worth considering personalizing product offers and marketing strategies for gender and age, particularly in categories like Beauty and Health, Electronics, and Clothing.
- Monitoring the stable Average Basket Value suggests consistent pricing and product range across demographics.

Card 6. Customer Demographics – Purchasing Preferences

- Payment methods by location.
- Payment methods by age group.
- Payment Method Trends Over Time
- Filters- Location,Year, Gender,Age Group,Quarter



Conclusions

Credit Card remains the dominant payment method in all locations, with a total of 22,096 transactions.

Largest locations by the number of transactions:

- Mumbai: 11,197 transactions
- Delhi: 10,799
- Bangalore: 8,249

- Hyderabad: 5,545

In all age groups, Credit Card is the most frequently used method. The 25–45 age group makes the most transactions across all methods (8.9 tys. for Credit Card alone).
Younger groups (18–25 years) are also active in alternative payment methods such as Google Pay UPI, PhonePe UPI, and Net Banking, although their total number of transactions is much lower (0.8–1.1 tys. per method).

Payment method trends over the year:

- Credit Card transactions are stable throughout the year, with slight increases in the second half (October–December).
- Debit Card, Net Banking, and UPI methods also maintain stable but much lower volumes.
- There are no strong seasonal spikes in alternative payment methods, indicating steady usage patterns.

Initial business diagnosis:

- Credit Card is the clear leader in payments, regardless of location and age. The company should continue optimizing its payment flows for credit cards.
- There is room for developing alternative methods (Net Banking, UPI, wallets), especially in younger groups and tech-savvy cities like Bangalore or Hyderabad.
- The second half of the year (October–December) shows a slight growth in payment activity, likely due to holiday season sales – this period could be supported by additional promotional campaigns.